User Service Database Design

1. Overview

Purpose

The User Service is responsible for managing Business Owner accounts on the KevaBook platform. The User Service maintains Business Owner account information and serves as the identity provider for authenticated platform users.

It provides functionality for:

- Business Owner registration
- Authentication
- Account management
- Profile management

NOTE: The User Service does not manage business profiles, services, bookings, availability, or notifications.

Indexes:

- Unique Index on email
- Index on status
- Index on created_at

Ownership Rules:

- User Service owns BusinessOwner data.
- Other services reference userId but do not create foreign key relationships.
- Business profiles, services, bookings, and notifications are managed by their respective microservices.

2. Database Technology

Property	Value
Database Type	MongoDb
Service	User service
Collection count	1
Primary Aggregate	Business owner

3. Aggregate Root

BusinessOwner

The BusinessOwner aggregate represents a registered platform user who can create and manage businesses on KevaBook.

4. Collection Design

Collection Name

business_owners

5. Document Structure

BusinessOwner Document

FIELD	TYPE	REQUIRED	DESCRIPTION
userId	String	Yes	Unique owner identifier
firstName	String	Yes	User first name

lastName	String	Yes	User last name
email	String	Yes	User email account
password	String	Yes	Encrypted password
status	Enum	Yes	Account state
emailedVerified	Boolean	Yes	Email verification state
createdAt	DateTime	Yes	Creation timestamp
UpdatedAt	DateTime	Yes	Last update timestamp

6. Account Status Enumeration

AccountStatus

Value	**DESCRIPTION**
ACTIVE	Account can access the platform
SUSPENDED	Account access is temporarily blocked
DEACTIVATED	Account closed and can't authenticate

7. Data Validation Rules

Field	**Validation**
firstName	Not null
lastName	Not null
email	Unique and valid email format
password	Required, and hashed
emailVerified	Boolean only

8. Ownership Rules

User Service Owns

BusinessOwner

User Service Does Not Own

The user service does not own Business, Service, Availability, Booking, Notification. These entities belong to their respective services.

9. Service Boundaries

The User Service acts as the system's identity provider for business owners.

External References

The User Service exposes only the **ownerId which is the userId field** to other services.

The Business Service stores ownerId but does not create a database foreign key because the data belongs to another microservice.

10. Supported Use Cases

UC-01 Register Account

Supported Fields are firstName, lastName, email, password, status

UC-02 Login

Supported Fields are email, password, status

Profile Management

Supported Fields are firstName, lastName, email

11. Supported API Operations

Endpoint	Purpose
POST /api/v1/users/registe	To register Business Owner
POST /api/v1/users/login	To authenticate Business Owner
GET /api/v1/users/userId}	To retrieve business owner profile
PUT /api/v1/users/{userId}	To update a profile
PATCH /api/v1/users/{ownerId}/password	To change password